higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

30 JULY 2019

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QUESTION 1

1.1 1.1.1 $$\lim_{x\to 0}\frac{arc\sin 4x}{arc\tan 5x}$$

L'hospital's rule since the condition $\frac{0}{0}$ holds; thus

$$\lim_{x\to 0}\frac{\frac{4}{\sqrt{1-16x^2}}}{\frac{5}{1+2x^2}}$$

$$\frac{4}{5}$$ (3)

1.1.2 $$\lim_{x\to 0}\sqrt{x}\ln x$$

$$= \lim_{x\to 0}\frac{\ln x}{\frac{1}{\sqrt{x}}}$$

L'hospital's rule since the condition $\frac{0}{0}$ holds; thus

$$\lim_{x\to 0}\frac{\frac{1}{x}}{\frac{-1}{2}x^{\frac{-3}{2}}}$$

$$\lim_{x\to 0}\frac{-2\sqrt{x^3}}{x}$$

L'hospital's rule since the condition $\frac{0}{0}$ holds; thus

$$= \lim_{x\to 0}\frac{-3\sqrt{x}}{1}$$

$$=0$$ (4)

1.2 $f(x)=sec\,x$

Thus, $f(x)$ is continuous at $x\neq\frac{\pi}{2}$ ✓and $x\neq\frac{3\pi}{2}$✓ i.e. $[0;\frac{\pi}{2})$ and $\left(\frac{\pi}{2};\frac{3\pi}{2}\right)$ (2)

[9]

QUESTION 2

2.1 $f(x)=\dfrac{x+1}{x+2}$

$$f'(x)=\lim_{h\to 0}\left[\frac{\frac{(x+h)+1}{(x+h)+2}-\frac{x+1}{x+2}}{h}\right]$$

$$¿\lim_{h\to 0}\left[\frac{\frac{(x+h)+1}{(x+h)+2}\times\frac{x+2}{x+2}-\frac{x+1}{x+2}\times\frac{(x+h)+2}{(x+h)+2}}{h}\right]$$

$$=\lim_{h\to 0}\left[\frac{\frac{x^2+xh+3x+2h+2-x^2-xh-3x-h-2}{(x+h+2)(x+2)}}{h}\right]$$

$$¿\lim_{h\to 0}\left[\frac{\frac{h}{(x+h+2)(x+2)}}{h}\right]$$

$$¿\frac{1}{(x+2)^2}$$ (5)

2.2 $y=arc\ \ cosec\ x$

$$cosec\ y=x$$

$$-cosec\ y\cot y\frac{dy}{dx}=1$$

$$\frac{dy}{dx}=\ -\frac{1}{cosec\ y\cot y}$$

$$=-\frac{1}{cosec\ x\sqrt{cosec^2x-1}}$$

$$\frac{-1}{x\sqrt{x^2-1}}$$ (3)

2.3 $y=\cot x$

$$y=\frac{\cos x}{\sin x}$$

$$\frac{dy}{dx}=\frac{\sin x(-\sin x)-\cos x(\cos x)}{\sin^2 x}$$

$$=\frac{-\sin^2 x-\cos^2 x}{\sin^2 x}$$

$$=-\frac{1}{\sin^2 x}$$ (3)

$$= -cosec^2 x$$

2.4 2.4.1 $$y=\sqrt{3x+\sqrt{2x+\sqrt{x}}}$$

$$\frac{dy}{dx}=\frac{1}{2}\left[3x+\left(2x+x^{\frac{1}{2}}\right)^{\frac{-1}{2}}\right]^{\frac{-1}{2}}\times\left[3+\left(2x+x^{\frac{1}{2}}\right)^{\frac{-1}{2}}\right]\times\left(2+\frac{1}{2}x^{\frac{-1}{2}}\right)$$

2.4.2 $$y=\left[\left(\frac{1}{x}+1\right)^{-1}+1\right]^{-1}$$

$$\frac{dy}{dx}=-\left[\left(\frac{1}{x}+1\right)^{-1}+1\right]^{-2}\times\left[-\left(\frac{1}{x}+1\right)^{-2}\right]\times\left(\frac{-1}{x^2}\right)$$

(2 × 3) (6)

2.5 $$y=(1-3x)^{\cos x}$$

$$\ln y=\cos x\ln(1-3x)$$

$$\frac{1}{y}\frac{dy}{dx}=\cos x\times\frac{1}{1-3x}\times-3-\sin x\times\ln(1-3x)$$

$$\frac{dy}{dx}=y\left[\frac{-3\cos x}{1-3x}-\sin x\ln(1-3x)\right]$$

¿ $$(1-3x)^{\cos x}\left[\frac{-3\cos x}{1-3x}-\sin x\ln(1-3x)\right]$$ (4)

2.6 $$e^{2x+3y}=x^2-\ln(xy^3)$$

$$e^{2x+3y}\left[2+3\frac{dy}{dx}\right]=2x-\frac{1}{xy^3}\left[y^3+x\times3y^2\frac{dy}{dx}\right]$$

$$2e^{2x+3y}+3e^{2x+3y}\frac{dy}{dx}=2x-\frac{1}{x}-\frac{3}{y}\frac{dy}{dx}$$

$$\left[3e^{2x+3y}+\frac{3}{y}\right]\frac{dy}{dx}=2x-\frac{1}{x}-2e^{2x+3y}$$

$$\frac{dy}{dx}=\frac{2x-\frac{1}{x}-2e^{2x+3y}}{3e^{2x+3y}+\frac{3}{y}}$$

(4)
[25]

QUESTION 3

3.1 3.1.1 $f(x)=2x^3-x^2+5$

$f'(x)=6x^2-2x=0$

$2x(3x-1)=0$

$x=0 \ \vee \ x=\frac{1}{3}$

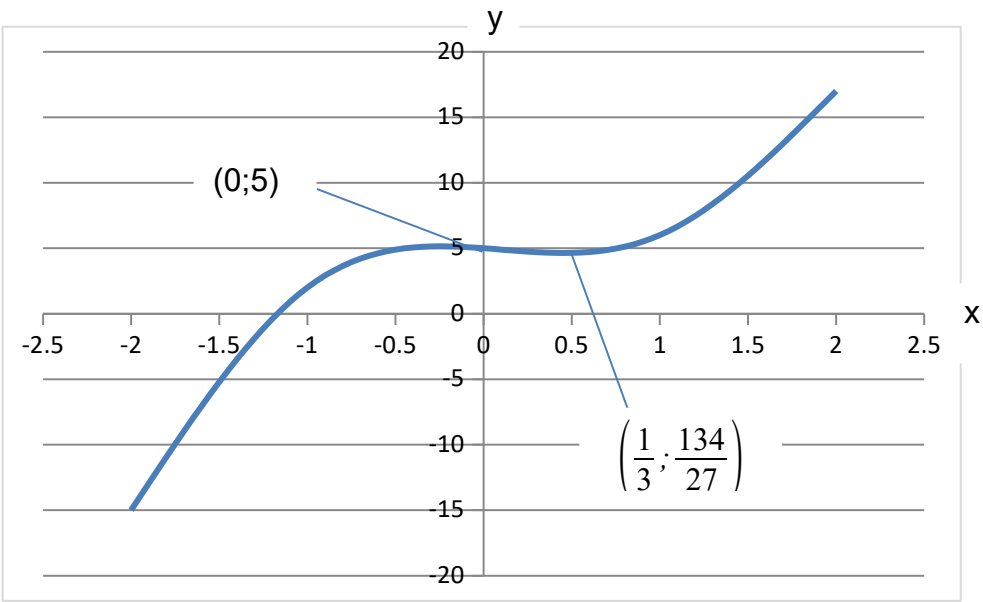
Turning points $\left(\frac{1}{3};\frac{134}{27}\right)$ and $(0;5)$ (2)

3.1.2

x	-2	-1	0	1	2
y	-15	2	5	6	17

(ONE mark for any 2 correct answers) (2)

3.1.3



(ONE mark for the shape)
(ONE mark for the indication of the point inflection on the graph) (2)

3.1.4

$$Let \quad x_0 = -1,5$$

$$f(-1,5) = -4$$

$$f'(-1,5) = 16,5$$

$$x_1 = -1,5 - \frac{(-4)}{16,5}$$

$$= -1,257575758$$

$$Let \quad x_1 = -1,257575758$$

$$f(-1,257575758) = -0,559200824$$

$$f'(-1,257575758) = 12,00413223$$

$$x_1 = -1,257575758 - \frac{(-0,559200824)}{12,00413223}$$

$$= -1,21099173$$

$$\approx -1,211 \qquad (4)$$

3.2

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dv}{dt} = \frac{4\pi}{3} \times 3r^2 \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{\frac{dv}{dt}}{4\pi r^2}$$

$$\frac{dr}{dt} = \frac{5 \ cm^3/s}{4\pi(10 \ cm)^2}$$

$$= 0,004 \ cm/s \qquad (4)$$

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QUESTION 4

4.1 4.1.1

$$\int 4\left(\frac{1}{x}-e^{-x}\right)\cos\left(e^{-x}+\ln x\right)\ dx$$

$$=4\int\left(\frac{1}{x}-e^{-x}\right)\cos\left(e^{-x}+\ln x\right)\ dx$$

$$let\ \ u=e^{-x}+\ln x$$
$$du=\left(\frac{1}{x}-e^{-x}\right)dx$$

$$=4\int\cos u\,du$$

$$¿\ 4\sin\left(e^{-x}+\ln x\right)+C \qquad (3)$$

4.1.2

$$\int\frac{1}{\sqrt{4-9x^2}}dx$$

$$¿\ \frac{1}{3}\int\frac{1}{\sqrt{\frac{4}{9}-x^2}}dx$$

$$¿\ \frac{1}{3}\sin^{-1}\left(\frac{3}{2}x\right)+C \qquad (3)$$

4.1.3

$$\int\frac{x^3+x}{x-1}\ dx$$

$$¿\int\left[x^2+x+2+\frac{2}{x-1}\right]dx$$

$$¿\ \frac{x^3}{3}+\frac{x^2}{2}+2x+2\ln(x-1)+C \qquad (5)$$

4.1.4

$$\int\sin 10x\sin 7x\,dx$$

$$¿\int\frac{1}{2}[\cos 3x-\cos 17x]dx$$

$$¿\ \frac{1}{2}\int(\cos 3x-\cos 17x)\,dx$$

$$¿\ \frac{1}{6}\sin 3x-\frac{1}{34}\sin 17x+C \qquad (3)$$

4.1.5 $$\int \frac{x}{sec\,3x}\,dx$$

$$¿ \int x\cos 3x\,dx$$

$let \quad u = x \qquad dv = \cos 3x\,dx$

$du = dx \qquad v = \frac{1}{3}\sin 3x$

$$\int u\,dv = uv - \int v\,du$$

$$¿ \frac{x}{3}\sin 3x - \frac{1}{3}\int \sin 3x\,dx$$

$$¿ \frac{x}{3}\sin 3x + \frac{1}{9}\cos 3x + C$$ (4)

4.2 $$\int \frac{x^2}{x^2 - 1}\,dx$$

$$\int \left[1 + \frac{1}{x^2 - 1}\right]dx$$

$$\frac{x^2}{(x+1)(x-1)} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$x^2 = A(x+1) + B(x-1)$$

$let \quad x = -1, \quad then \quad B = -\frac{1}{2}$

$x = 1, \quad then \quad A = \frac{1}{2}$

$$¿ \int dx + \frac{1}{2}\int \frac{1}{x-1}\,dx - \frac{1}{2}\int \frac{1}{x+1}\,dx$$

$$= x + \frac{1}{2}\ln(x-1) - \frac{1}{2}\ln(x+1) + C$$ (6)

[24]

QUESTION 5

5.1 $\int_0^{\infty} e^{-st}.f(t)\,dt$

$=-5\int_0^{\infty} e^{-st}\,dt$

$\therefore -5\left[\frac{-e^{-st}}{s}\right]_0^{\infty}$

$=-5\left(0+\frac{1}{s}\right)$

$\therefore \frac{-5}{s}$ (4)

5.2 5.2.1 $3x^2=6x$

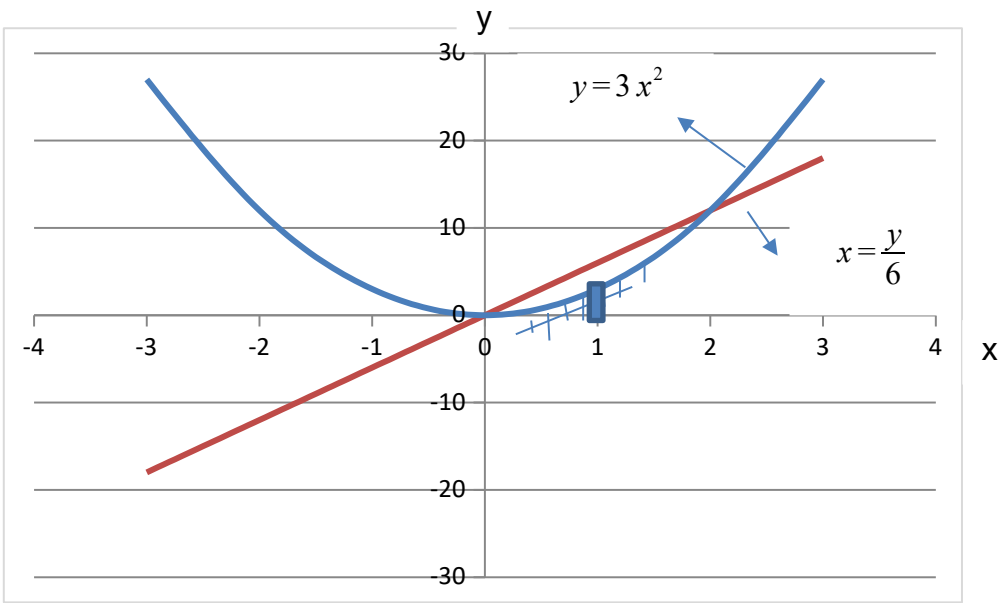
$3x^2-6x=0$

$3x(x-2)=0$

$x=0 \vee x=2$

Thus the coordinates of the point of intersection are $(0;0)$ and $(2;12)$ (2)

5.2.2



(ONE mark for indication of enclosed area)
(ONE mark for indication of the vertical or vertical strip) (2)

5.2.3 $A_x = \int_a^b (y_1 - y_2)\,dx$

$\therefore \int_0^2 (6x - 3x^2)\,dx$

$\therefore [3x^2 - x^3]_0^2$

$= 4 \ units^2$ (3)

5.2.4 $V_x = \int_a^b (y^2_1 - y^2_2)\,dx$

$\therefore \int_0^2 (36x^2 - 9x^4)\,dx$

$\therefore \left[12x^3 + \frac{9}{5}x^5\right]_0^2$

$= 38{,}4 \ units^3$ (4)

5.3 $(d)_y = r^2 A$

$(d)_y = r^2\,dA$

$(d)_y = x^2 b\,dx$

$= b\int_{\frac{-a}{2}}^{\frac{a}{2}} x^2\,dx$

$= b\,a^2\left[\frac{-x^3}{3}\right]_{\frac{-a}{2}}^{\frac{a}{2}}$

$= b\left[\frac{a^3}{12}\right]$

$\therefore \frac{a^2 A}{12}$ (4)

[19]

QUESTION 6

6.1 $\left(3y^2+2\right)\cos x\,dx - 6y\sin x\,dy = 0$

$$\frac{6y}{3y^2+2}\,dy = \cot x\,dx$$

$$\ln\left(3y^2+2\right) = \ln\left(\sin x\right) + C \qquad (3)$$

6.2 $$\frac{d^2y}{dx^2} = 3x^2 - 2x$$

$$\frac{dy}{dx} = x^3 - x^2 + A$$

$$A = -1$$

$$\frac{dy}{dx} = x^3 - x^2 - 1$$

$$y = \frac{1}{4}x^4 - \frac{1}{3}x^3 - x + B$$

$$B = \frac{25}{12}$$

$$y = \frac{1}{4}x^4 - \frac{1}{3}x^3 - x + \frac{25}{12} \qquad (6)$$

[9]

TOTAL: 100